Laboratory 9

Stopwatch Datapath

9.1 Outcomes and Objectives

The outcome of this lab is to design and simulate a datapath to store and manipulate the information needed to run a stopwatch. Through this process you will achieve the following learning objectives:

- Design a circuit that performs register transfer
- Datapath and Control Architecture
- Design a control word table and specify the control word values for every state.
- Creating a Do file to automate waveform setup

9.2 Stopwatch

A stopwatch is a device that is used to measure time intervals, usually in competitive events. The stopwatch that you will be designing gets its input from two buttons. The stopwatch will measure accurately to a 1/10th of a second. The time will be displayed using 3 digits which will represent tenths of a second, unit second, and tens of seconds. As a result, the stopwatch is limited to measuring intervals of time from 0.1 second to 99.9 seconds.

The stopwatch's behavior is dictated by its 2 buttons, called `S1` and `S2`, according to the finite state machine (FSM) shown in Figure 9.1.

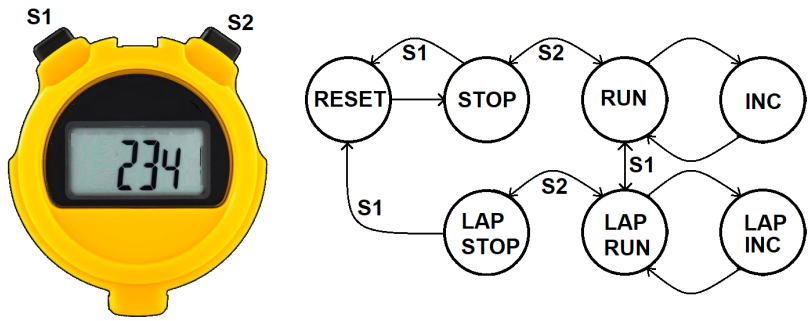


Figure 9.1: A digital stopwatch gets its input from 2 buttons and displays its output on a 7-segment display. The behavior of the stopwatch can be described by this FSM.

To make sense of the FSM shown in Figure 9.1, it's helpful to imagine timing a 4-person relay race. In this race, each athlete runs one lap and then passes a baton to the next runner. The time required for a runner to complete one lap is called their split time. In order to measure each runner's split time, you need to be able to stop the displayed time while allowing the stopwatch to continue to run its internal timer. This is called a lap feature. Let's explore how the lap feature works by timing the mile relay at a Mines track meet. As we go through this scenario, reference the finite state machine in Figure 9.1.

Prior to the start of the race, you push the `S1` button, putting the stopwatch into the `RESET` state. This clears the internal timer and the displayed time. The stopwatch automatically goes back to the `STOP` state. You are ready for the start of the race. The gun goes off and you immediately press the `S2` button, putting the stopwatch into the `RUN` state. The internal timer is keeping track of the elapsed time and you see the displayed time changing to reflect the internal timer. As soon as the first runner hands the baton to the second runner, you press the `S1` button, putting the stopwatch into the `LAP RUN` state. This causes the displayed time to stop, showing the time at the instant you pressed `S1`, while simultaneously allowing the internal timer to keep running. The internal timer is now keeping track of the elapsed time since the start of the race. You calmy write down the displayed time on your clipboard (made easier because it is not changing) and then press `S1` again, putting the stopwatch back into the `RUN` state. You repeat this process until the last runner comes in. As soon as they do, you press the `S2` button, stopping the internal timer and showing the time the last runner crossed the finish line.

In this scenario, we did not put the stopwatch into the `LAP STOP` state – this state would stop the internal timer and keep the displayed time the moment the `S1` button was pressed when the stopwatch was in the `RUN` state. You should also note, we did not talk about the `INC` and `LAP INC` states. Let's explore them now.

Clearly, the stopwatch does not increment the stored and displayed time at 50Mhz. The stored and displayed time counts up every 10th of a second. This is managed by the `counter/comparator` at the top of Figure 9.2 which asserts the **tenth** signal every 10th of a second. This signal tells the FSM in Figure 9.1 to transition from the `RUN` state to the `INC` state. In the INC state the stored time is incremented and the counter at the top of Figure 9.2 is reset back to `0`. The `LAP INC` state performs a similar function for the `LAP` state.

To summarize:

- `RESET` – Reset the internal and displayed time values.

- `STOP` – Stop the 10th second timer and hold the displayed time.
- `RUN` – Run the 10th second timer and display the stored time.
- `INC` – Increment the stored time
- `LAP RUN` – Run the 10th second timer and hold the displayed time.
- `LAP INC` - Increment the stored time.
- `LAP STOP` – Stop the 10th second timer and hold the displayed time.

9.3 Module: stopWatchDatapath

The datapath for the stopwatch is shown in Figure 9.2. The behavior of the datapath to perform the functions of a stopwatch follows.

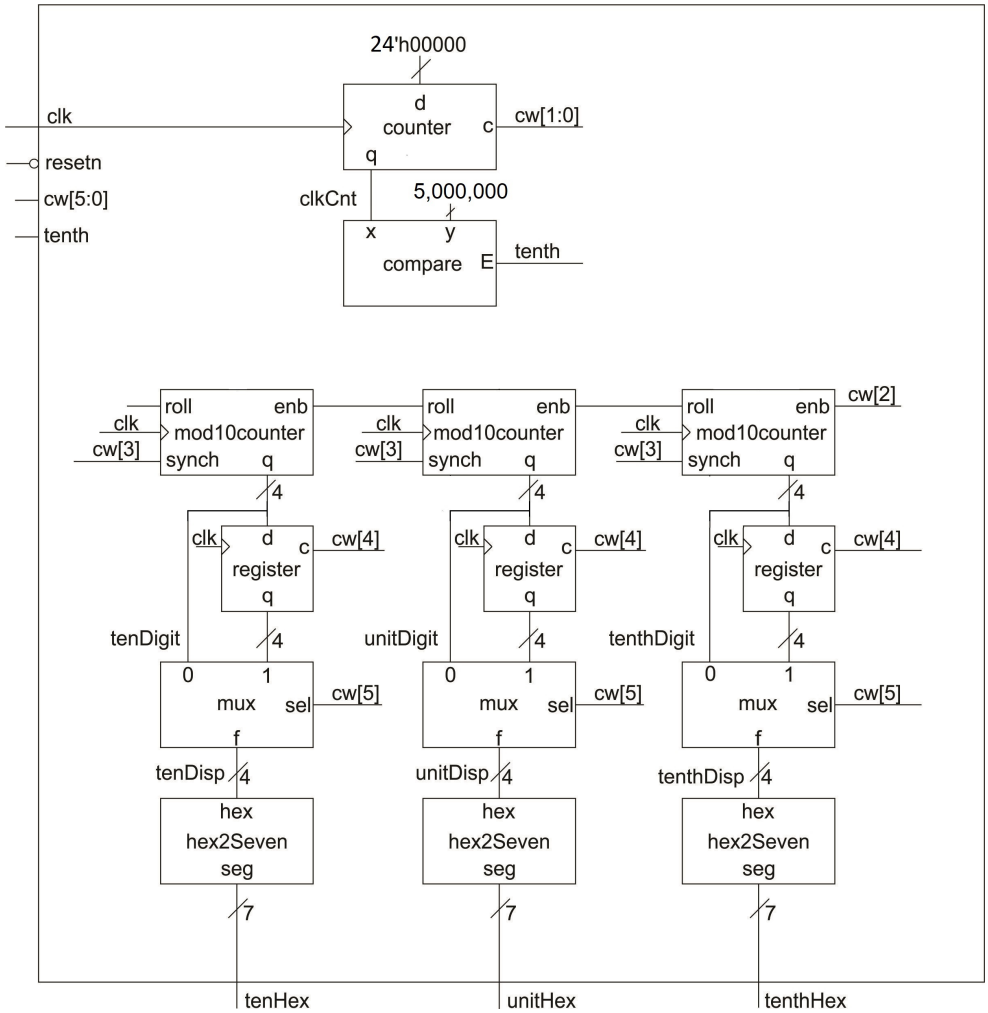


Figure 9.2: The datapath for the stopwatch has a 6-bit control word and displays the time on three 7-segment displays.

The internal time is formed by the `counter/comparator` combination at the top of Fig-

ure 9.2. The `clk` input from the Development Board runs at 50MHz. This will be the rate at which the `clkCnt` output from counter counts up. When `clkCnt` counts from `0` to `5,000,000` then $1/10^{th}$ of a second has gone by. The `roll` signal indicates when $1/10^{th}$ of a second has gone by. The control unit you will build in the next lab, will use this signal to coordinate incrementing the bank of mod 10 counters in the datapath. Note that the bank of mod 10 counters are incremented in a state called `INC`, that is not shown in Figure 9.1. The output of the mod10 counters, `tenDigit`, `unitDigit` and `tenthDigit` can be latched-up in the register bank so that the datapath can hold the displayed time while still allowing the bank of mod 10 counters to keep track of the elapsed time. The multiplexer controls what is displayed. The 4-bit digit representing a time digit, `tenDisp`, `unitDisp` and `tenthDisp` are converted into a 7-segment pattern before leaving the datapath.

To better understand the datapath, construct the control word for each state. In order to help you, the actions associated with each state are outlined below. In some cases, you will need to make decisions about the control bits. Use your understanding of the datapath's operation and your intuition of how you would want to the datapath to work. If this is uncomfortable, please understand that it's important for you to learn how to make design decisions on your own so that you can reach your full potential as an engineer.

- `RESET` – clear the values in the registers and counters
- `STOP` – hold the timer counter and display the mod 10 counters
- `RUN` – allow the timer counter to count up and display the mod 10 counters
- `INC` – clear the timer counter and increment the mod 10 counters
- `RUN2LAP` – Latch up the mod 10 counters in the lap register
- `LAP RUN` – allow the timer counter to count up, display the latched time
- `LAP INC` – clear the timer counter and increment the mod 10 counters
- `LAP STOP` – stop the timer counter and display the latched time

Table 9.1: Control word table for the datapath shown in Figure 9.2.

	cw[5] **2x1 mux**	**cw[4]** **lap register**	**cw[3]** **mod10 reset**	**cw[2]** **mod10 count**	**cw[1:0]** **timer counter**
	0 = mod10	1 = load	1 = reset	1 = count up	11 = load
	1 = register	0 = hold	0 = hold	0 = hold	10 = count up
					01 = not used
					00 = hold
`RESET`			1		
`STOP`					
`RUN`					
`INC`					
`RUN2LAP`	1			0	
`LAP RUN`					10
`LAP INC`		0			
`LAP STOP`	1				

Now that you have the control word figured out, you need to write the Verilog code for the datapath. For the datapath module:

- Use the datapath.v file provided in the Canvas folder as the starting point.
  - Use the module definitions from previous lab and the Canvas lab folder for this lab.
- Provide meaningful names to the wires in the module.
- Properly tab-indent your code.

- Single level for wire declarations
- Single level for component instantiations
- Two levels for case statement
- Three levels for case values

9.4 Testbench

Before you download your completed datapath to the Development Board, you are going to perform extensive simulations to uncover as many bugs as possible. Trust me, errors are much, much easier to find in a simulation.

There is a practical consideration that will make the simulation more manageable. The `counter-comparator` combination in Figure 9.2 acts as a clock divider circuit. The counter counts up at 50MHz (the main oscillator frequency). The comparator checks when the count reaches 5,000,000 = `0x4C4B40`, meaning that a $1/10^{\text{th}}$ of a second has gone by. You do not want to have to run the counter to 5,000,000 in your simulation, it would just take too long. To replace the 5,000,000 constant you need to go into the datapath.v module and modify the following:

- Set `localparam tenthSecondConstant` to 4'h000002 (an arbitrary small constant)
- Set the parameter `N` to 4, this sets the word size of the counter/comparator hardware.

You can use the parameter `N` as the generic parameter in component instantiation for generic components. For example, you will have the following `genericCounter` instantiation in your datapath - the use of the parameter value `#(N)` as the width of the counter is legal syntax in Verilog as long as `N` is defined somewhere.

```
genericCounter #(N) tenthSecondCounter(clk, resetn, zero24, cw{[}1:0{]}, clkCount);
```

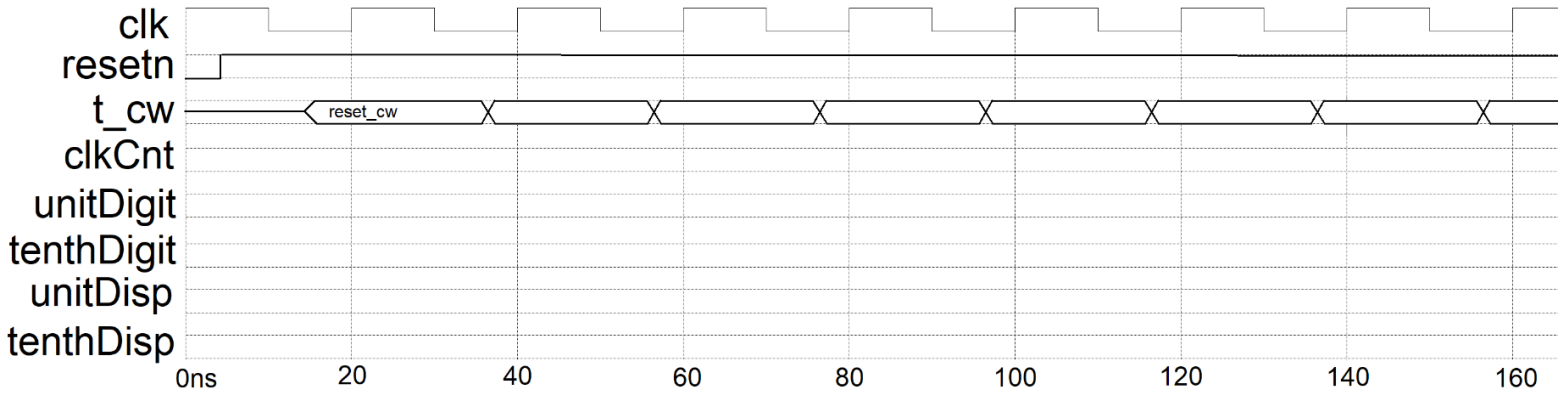
Make a note to yourself to modify `N` and `tenthSecondConstant` values before synthesis. This is also mentioned later in this document.

Next modify the control words defined in the datapathLab09_tb.v file using the values from Table 9.1.

Finally, you need to understand what output the simulation should output so that you can compare that to what your simulation actually producing. Any difference between these two indicate an error (either in your understanding or circuit behavior) that need to be fixed.

To do this, complete the entries in the following timing diagram figures. Use the code in the testbench to figure out the values for the control word and how long they are held. The control word does not necessarily change every 20ns. When it does not, you can just rewrite the control word or edit the image to connect adjacent cells.

The $display statement in the testbench prints out a message when that line of the simulation is reached. Including these statements helps you to understand where your simulation is at and what behavior to expect.


clk
resetn
t_cw
clkCnt
unitDigit
tenthDigit
unitDisp
tenthDisp
reset_cw
0ns
20
40
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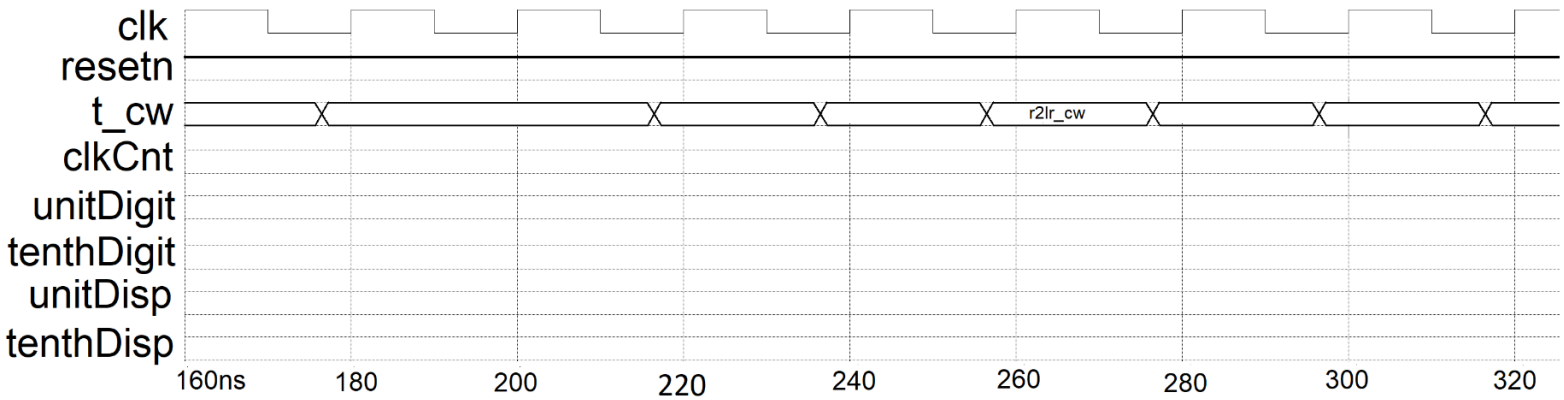
100
120
140
160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resetn
t_cw
clkCnt
unitDigit
tenthDigit
unitDisp
tenthDisp
r2lr_cw
160ns
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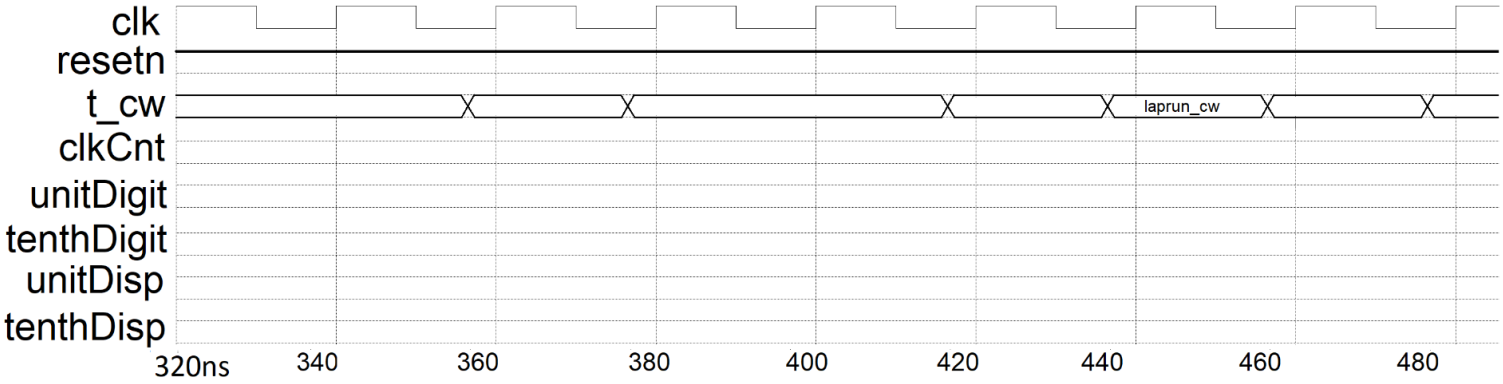
200
220
240
260
280
300
320


clk
resetn
t_cw
clkCnt
unitDigit
tenthDigit
unitDisp
tenthDisp
laprun_cw
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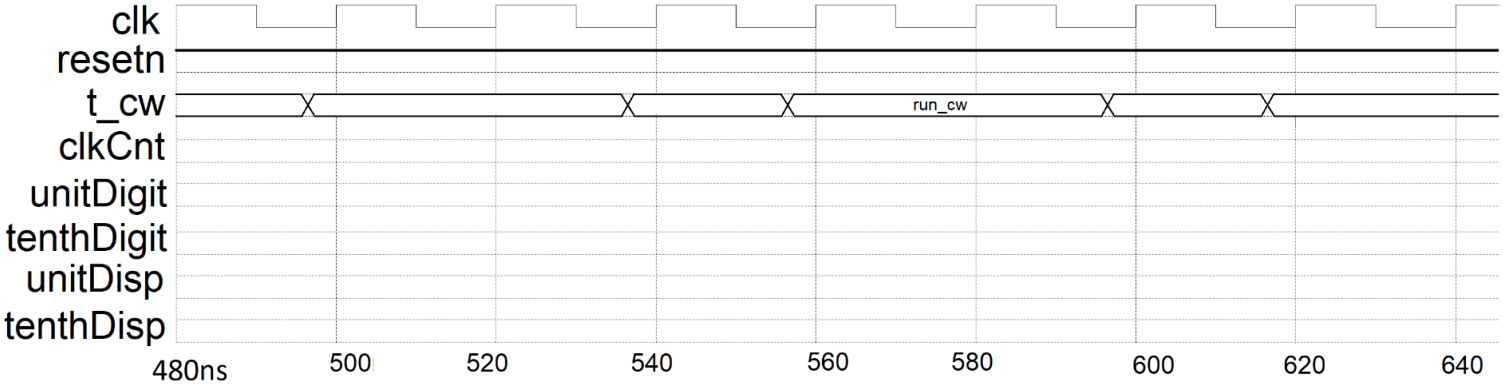
360
380
400
420
440
460
480


clk
resetn
t_cw
clkCnt
unitDigit
tenthDigit
unitDisp
tenthDisp
run_cw
480ns
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520
540
560
580
600
620
640

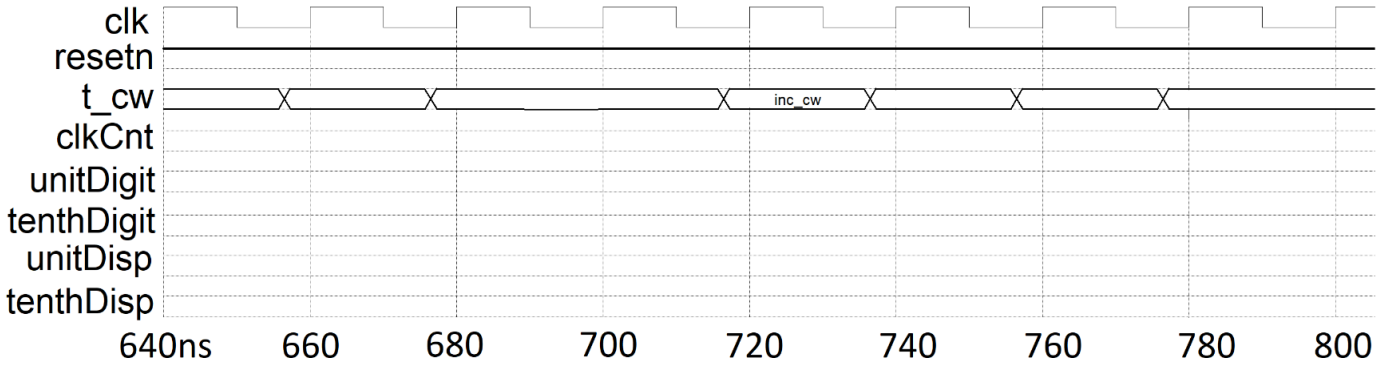


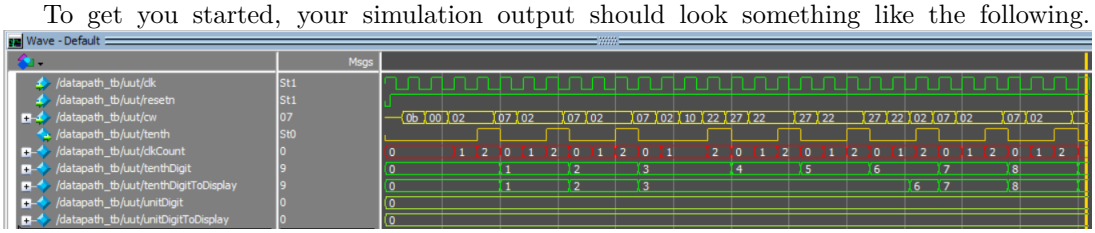
Figure 9.3: Complete the timing diagram using the control words found in the testbench.

Before running the simulation, you need to complete the do file given to you on Canvas. Organize the waveforms on your simulation as follows.

signal	radix	trace color
`clk`	default	green
`resetn`	default	green
`cw`	hex	yellow
`tenth`	default	gold
`clkCnt`	hex	red
`tenthDigit`	hex	green
`tenthDisp`	hex	green
`unitDigit`	hex	greenyellow
`unitDisp`	hex	greenyellow

Once you have the do file complete, launch the testbench simulation in Questa and compare the expected output from Figure 9.3 against the actual output from your simulation. Since you will not be synthesizing your design, you need to ensure **perfect** functionality as you will use this module in an upcoming lab. Do this by looking for discrepancies starting at time 0 and only advancing the simulation when everything is correct. You will have to transcend the design hierarchy to find the source of your errors. Most errors are due to incorrect wiring of modules – wrong signal names or wrong signal order.

To get you started, your simulation output should look something like the following.



9.5 Turn in

You may work in teams of at most two. Make a record of your response to the items below and turn them in a single copy as your team's solution on Canvas using the instructions posted there. Include the names of both team members at the top of your solutions. Use complete English sentences to introduce what each of the following listed items (below) is and how it was derived. In addition to this submission, you will be expected to demonstrate your circuit at the beginning of your lab section next week.

System Architecture

- Complete Table 9.1.
- Verilog code for the body of the datapath module (courier 8-point font single spaced), leave out header comments.

Testbench

- Complete timing diagram from Figure 9.3.
- A copy of your simulation timing diagram. Use the signal color and order specified.
- Demonstrate that your simulation to a member of the lab team.